Final ML Project

Income Prediction Using Machine Learning

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Predictive Goals

- **Problem Type:** Binary Classification
- **Dataset:** Adult Census dataset (32,000+ records)
- **Target:** 0 → <= $50K
  1 → > $50K
- **Features:** 14 distinct variables including education, occupation, and capital gains.

The Pipeline

Preprocessing Steps:

- Handled missing values ("?")
- One-hot encoding
- Converted target to 0/1

Split:

- Train: 70%
- Validation: 15%
- Test: 15%

Final Comparison Table (Validation & Test Results)

	Val Accuracy	Val Precision	Val Recall	Val F1	Val ROC-AUC	Test Accuracy	Test Precision	Test Recall	Test F1	Test ROC-AUC
Logistic Regression	0.855446	0.736419	0.622449	0.674654	0.909426	0.851996	0.736184	0.600340	0.661358	0.906722
Decision Tree	0.812654	0.607407	0.627551	0.617315	0.749455	0.811259	0.602254	0.636054	0.618693	0.751432
Random Forest	0.856675	0.734714	0.633503	0.680365	0.907709	0.856909	0.742132	0.621599	0.676539	0.907104
Gradient Boosting	0.869165	0.792165	0.619048	0.694988	0.924814	0.867758	0.790570	0.613095	0.690613	0.923070
KNN	0.824324	0.650568	0.584184	0.615591	0.846971	0.823337	0.649475	0.578231	0.611786	0.845085
SVC	0.858313	0.763617	0.596088	0.669532	0.900576	0.847083	0.739130	0.563776	0.639653	0.892816
Voting Ensemble	NaN	NaN	NaN	NaN	NaN	0.866530	0.786026	0.612245	0.688337	0.919579
Naive Bayes	NaN	NaN	NaN	NaN	NaN	0.797544	0.667864	0.316327	0.429313	0.831800

Conclusion

Best Model: **Gradient Boosting**

Why:

- Highest F1-score
- Strong ROC-AUC
- Best generalization

Takeaways:

- Ensemble models perform well
- Tree-based models outperform others